

Perspectives on AI, Business and Sustainability PAIBS

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The Space Sector

EO Use Cases

AI4EO

Project Definition

Earth Observation (EO) Data

Ground Reference Data

TeamWork Time

1. Frame real-world challenges

- **Identify** and **define complex problems** in business and sustainability.
- Align projects with the UN Sustainable Development Goals (**SDGs**).
- Gather **stakeholder needs** through **qualitative research & user insights**.

2. Leverage Earth Observation & AI

- Understand the fundamentals of the space sector and EO data.
- Apply AI methods to extract insights and **create solutions** from EO data.
- Explore opportunities for innovation and disruption **across industries**.

3. Design and prototype solutions

- Use **design thinking** to generate and evaluate ideas.
- **Develop and prototype** AI-driven services or applications.
- Integrate considerations of **feasibility, scalability, and sustainability**.

4. Manage projects with professional methods

- Work effectively in **multidisciplinary teams**.
- Apply tools such as Git for version control and documentation.
- Practice agile project management techniques.

5. Assess impacts and communicate effectively

- Evaluate the **social, ethical, and economic implications** of AI solutions.
- **Communicate** results to both **technical and business audiences**.
- **Prepare for the Final Degree Project** with improved technical and professional literacy.

Term 1 (October 8th - December 3rd) → *From challenge to solution concept*

Definition and Solution Design - identifying meaningful challenges, understanding stakeholder needs, and proposing comprehensive solution.

Term 2 (February 12nd - May 14th) → *From concept to working prototype*

Technical Implementation - technical development, culminating in a fully developed prototype and business case.

Introduction: Sessions Term 1

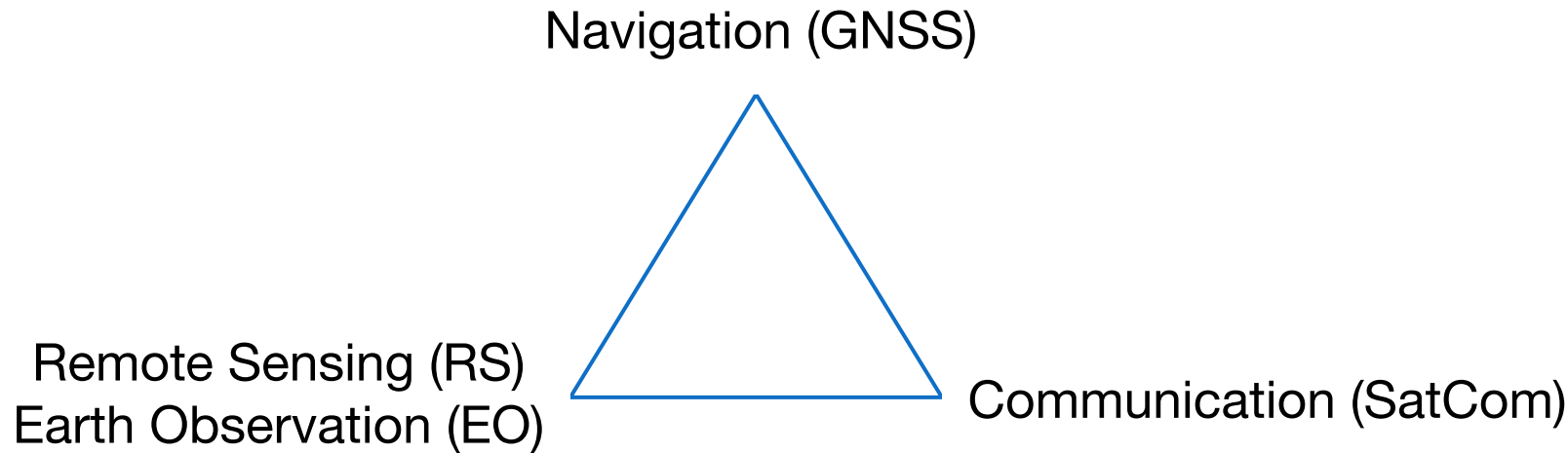
Term 1		
Date	Content	Teacher/s
October 8, 2025	Challenge Introduction	Marc Herrera / Angelo Romasanta
October 22, 2025	Systems Thinking	Guillermo Casasnovas
November 5, 2025	Design Thinking	Aurélie Glorieux
November 12, 2025	User Analysis, Personas, and Journey Mapping	Aurélie Glorieux
December 3, 2025	Midterm Presentations	Marc Herrera / Angelo Romasanta

Term 2		
Date	Content	Teacher/s
February 12, 2026	Literature Replication	Marc Herrera
February 19, 2026	Prototyping tools and techniques	Jose Rodriguez
March 12, 2026	Front-end prototyping	Jose Rodriguez
April 9, 2026	Prototyping AI applications	Jose Rodriguez
April 16, 2026	AI applications in Earth Observation	Marc Herrera
April 30, 2026	Impact assessment	Angelo Romasanta
May 14, 2026	Final Presentations	Marc Herrera / Angelo Romasanta

Component	Weight	Key Points
Group Project	60%	Problem definition & research (20%) - Technical implementation (50%) - Business validation (20%) - Communication (10%) - Peer review multiplier (x0.8–1.2)
Individual Tasks	40%	- Midterm deliverable (15%) - Final exam (25%)

Milestones:

- Midterm presentation (Session 5)
- Final presentation (Session 12)
- 2 mandatory coaching sessions (Terms 1 & 2)



- The space sector is no longer only **governments** → **commercial players** now drive innovation.
- Global market size in **2023** exceeded **\$500 billion**, projected to surpass **\$1 trillion by 2040** (Morgan Stanley / PwC).
- "**NewSpace**": a term for new commercial initiatives — think SpaceX, Planet, Rocket Lab, Starlink.
- Beyond satellites: includes launch providers, ground stations, data analytics companies, AI-driven startups.

Space = infrastructure + services, like roads or internet → enabling countless industries.

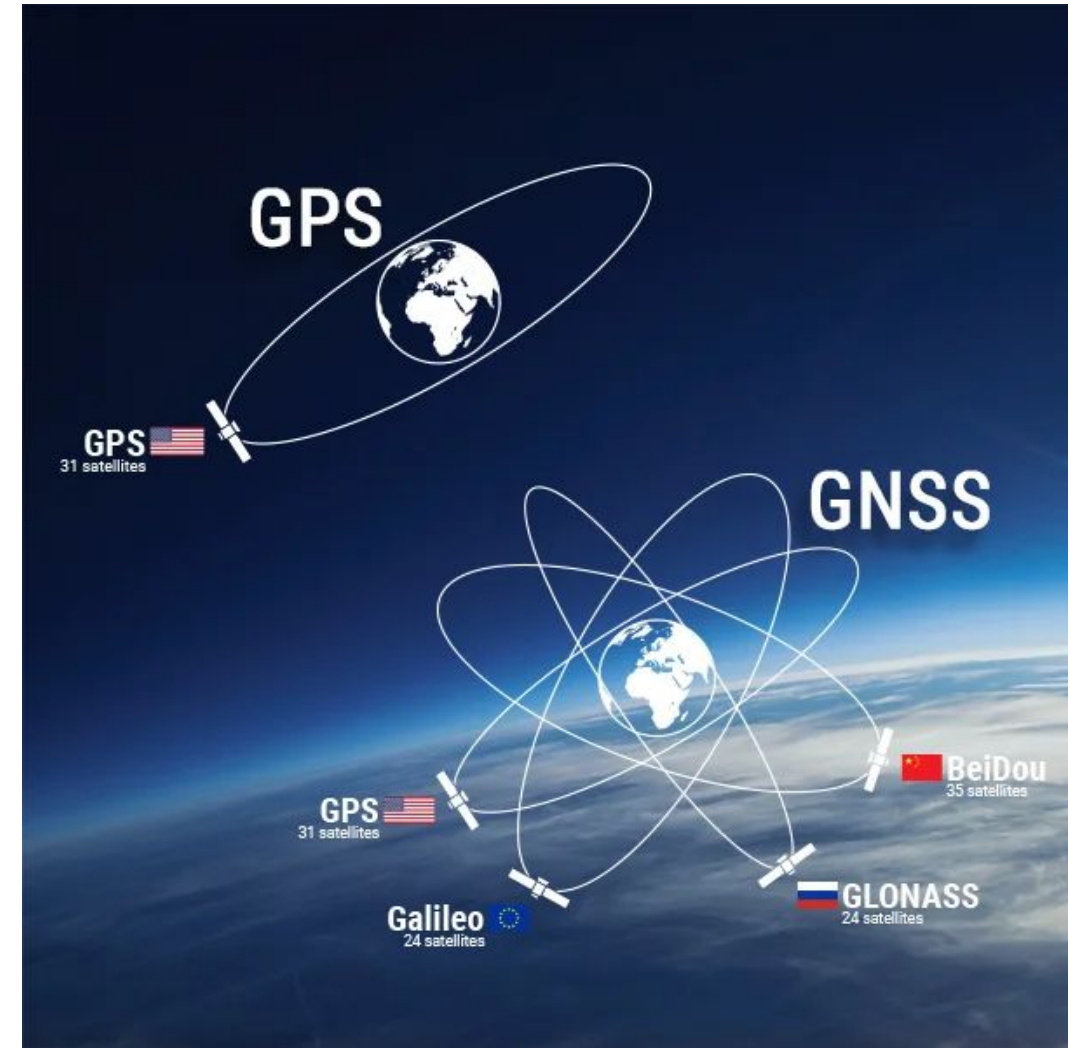
GNSS, *Global Navigation Satellite Systems*, underpin transport & logistics.

Main systems:

- GPS (US, ~1993 full ops)
- Galileo (EU, newer, more precise)
- BeiDou (China, full since 2020)
- GLONASS (Russia)

Uses:

- Finance: stock exchanges use GNSS timing for transactions (microsecond synchronization).
- Everyday life: ride-hailing (Uber), smartphone maps, fitness trackers.
- Emerging uses: drones, precision farming, autonomous vehicles, IoT sensors.
- Business aspect: Galileo is the EU's first major infrastructure project → sovereignty over navigation services.



Communications (SatCom)

Traditional SatCom: TV & radio broadcasting, military communications. Later regular comms.

Now:

- Mega-constellations (e.g., Starlink with >6,000 satellites)
→ Global **internet**: planes, cruise ships, rural areas.
- SatCom critical in disaster relief when ground networks fail (earthquakes, floods).
- Maritime & aviation: allows real-time data, fuel optimization, safety tracking.

Impact:

Business: SatCom enables inclusion → connecting billions without internet.

Geopolitics: broadband constellations influence digital sovereignty.

From broadcast-only to two-way, low-latency internet → direct competition with terrestrial providers

Remote Sensing (RS) / Earth Observation (EO)

Remote Sensing (RS) is the **collection of data** about the Earth's surface without direct contact.

RS allowed us to **monitor and study our planet** in new ways, providing a global view of the Earth and enables observation of changes over time.

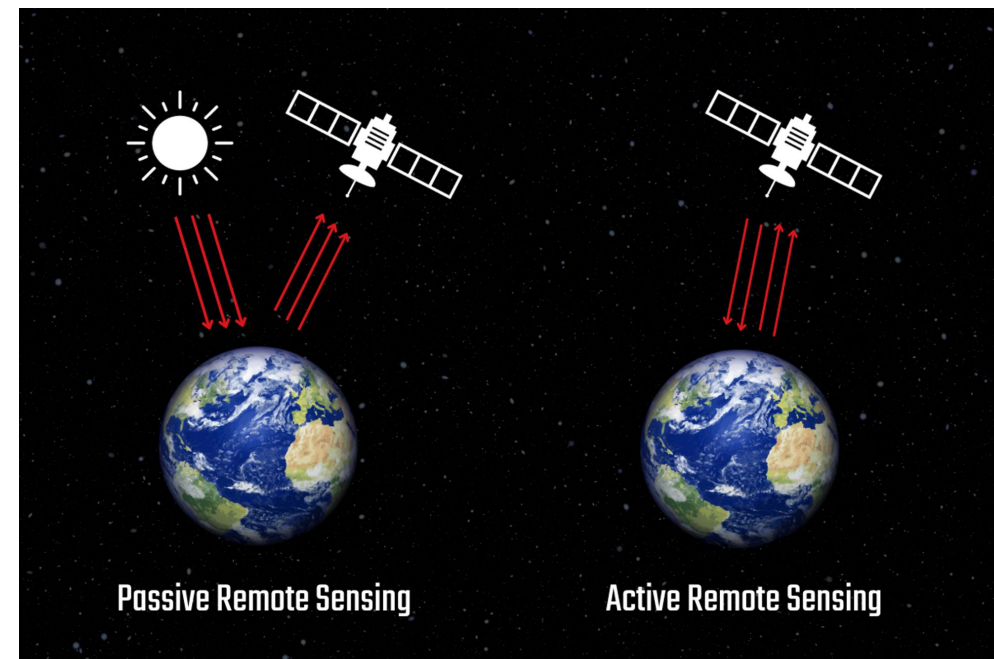
EO **combines data** from multiple sources to create comprehensive views of the planet.

RS provides the data — **EO** turns it into **understanding**.

The EO space sector hosts a wide range of use cases, business models, and emerging companies growing each year.

EO value chain:

- Upstream (hardware)
- Downstream (data = services)



Remote Sensing: Upstream

Upstream: *Looking up*, the infrastructure side of EO.

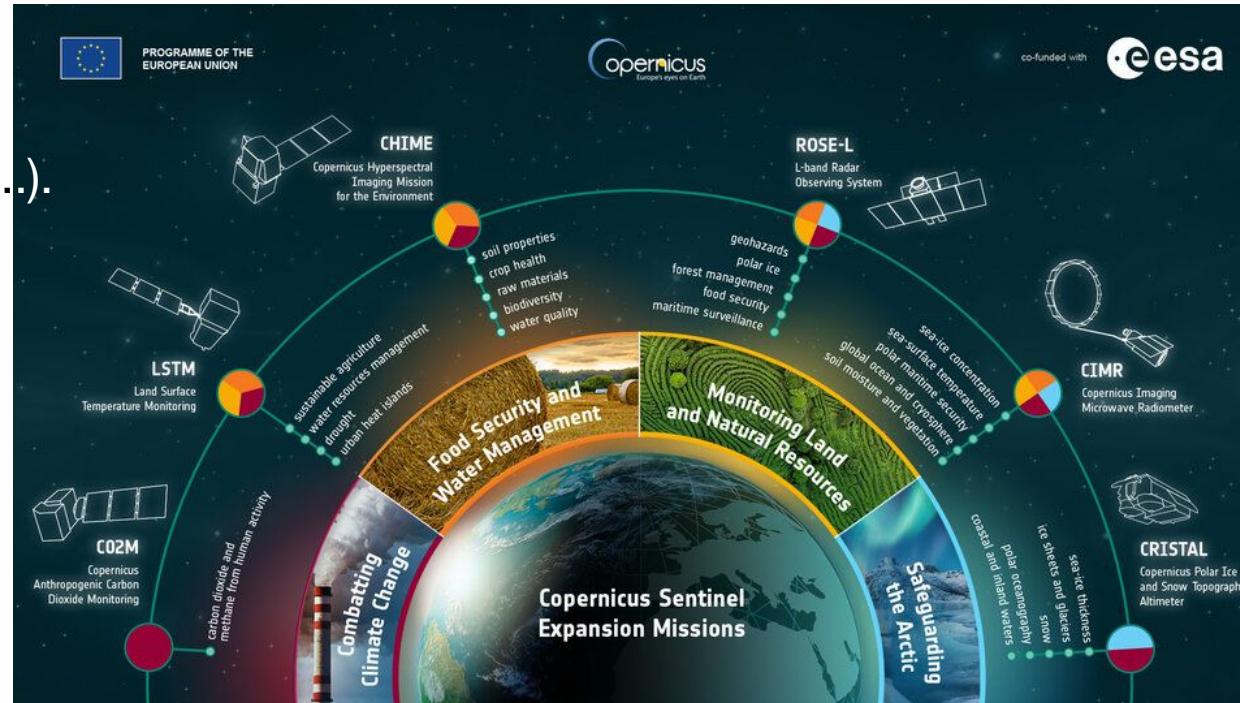
Not only Public but also private sector involvement: ESA (Copernicus), NASA (Landsat), commercial constellations (Planet, Maxar).

Barriers to entry are high → focus on engineering, capital, regulation.

Sets the foundation for downstream services — without upstream, no EO data.

Includes

- Satellite platforms.
- Sensors (optical, radar, ...).
- Launch systems.



Downstream: *Looking down*, data services & applications built on EO data.

MUSTS

- Accessible for business innovation and AI-based solutions.
- Generates real-world value: improves decision-making, enables sustainability initiatives.
- Demonstrates the space-to-society value chain: infrastructure → data → actionable insights.

Barriers to entry are low → open access data, direct implementation, cloud platforms.



The projects will be on the downstream: using EO + AI to solve complex problems.

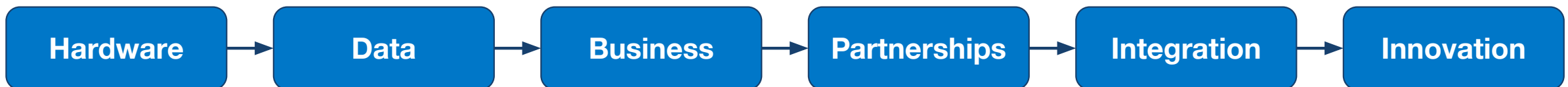
Space sector shift from **hardware** (rockets, satellites) to **data**.

Example: Copernicus (EU) provides free EO data → **enables** new business models.

New business models → **generates** new startups (agriculture advisory, carbon monitoring, ESG services).

Result:

- **Public-private partnerships** (PPPs): governments fund infrastructure, companies build services (ESA + Airbus, NASA + SpaceX).
- **New type of Integration:** AI + EO + IoT (sensors on Earth) → smarter solutions.
- **Innovation:** EO data = raw material, and AI = engine to transform it into products/services.



The Future of the Space Sector

Space Exploration (Moon, Mars, beyond)

Renewed race: NASA's Artemis program aims for a permanent Moon base by 2030.

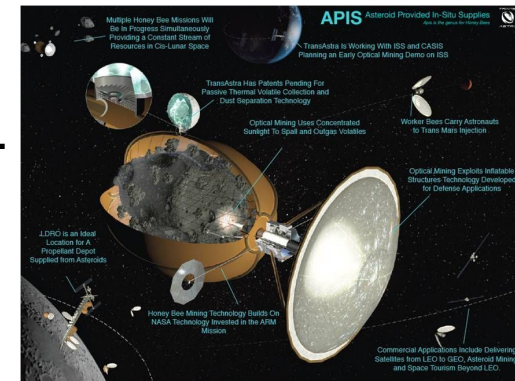
Mars ambitions: SpaceX's Starship designed for interplanetary transport.



Space Mining (asteroids, lunar resources)

Asteroids and the Moon hold vast amounts of metals (platinum, rare earths, water ice).

Potential to fuel rockets with lunar water → reduces cost of deep space missions.



Space Tourism (suborbital, orbital stays)

First wave is suborbital flights (Virgin Galactic, Blue Origin).

Orbital tourism: SpaceX Crew Dragon has already flown private astronauts.

Prices falling (though still >\$200k per seat), opening future markets.



Orbital Habitats (private space stations)

ISS is aging → private replacements on the way.

Axiom Space building commercial station modules.

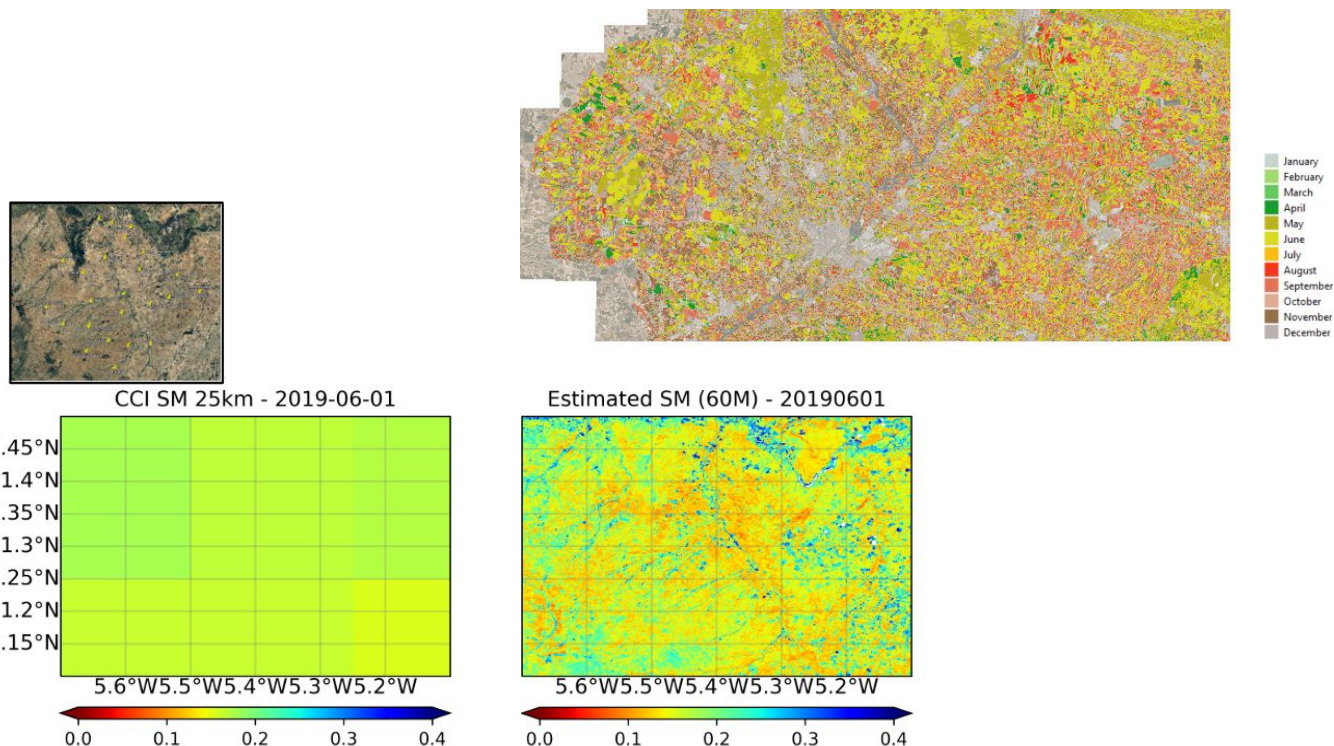


VAST Space developing inflatable orbital habitats → first private space station around 2025–2030.

Soil and Agriculture

Management of agricultural areas, predicting and explaining crops phenological status, crop classification and detection, soil moisture, soil displacement, thermal emissions, generation of cover maps, managing food resources, detecting herds.

→ SDGs: 2 (Zero Hunger), 12 (Responsible Consumption & Production), 13 (Climate Action), 15 (Life on Land)



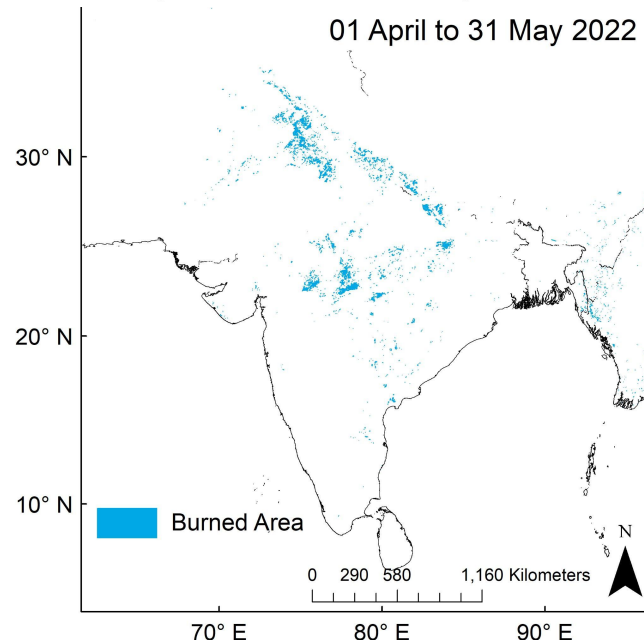
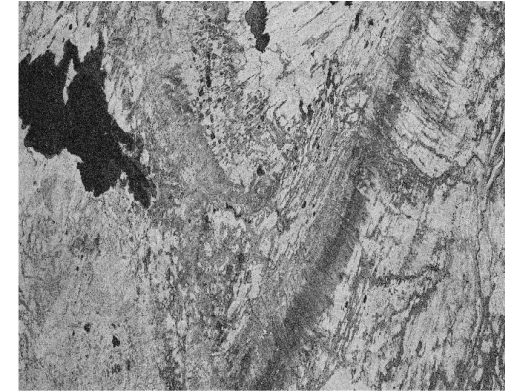
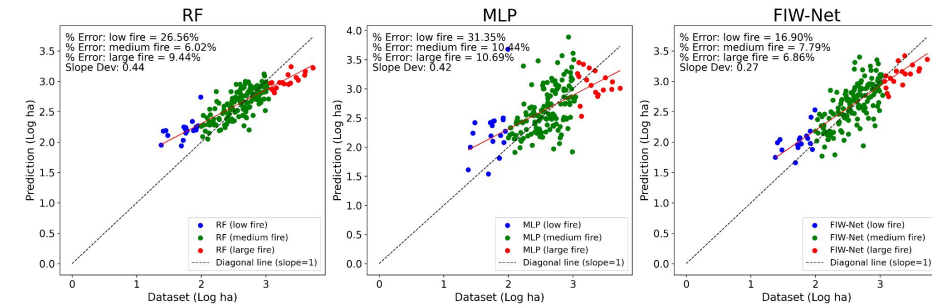
SUSTAINABLE DEVELOPMENT GOALS



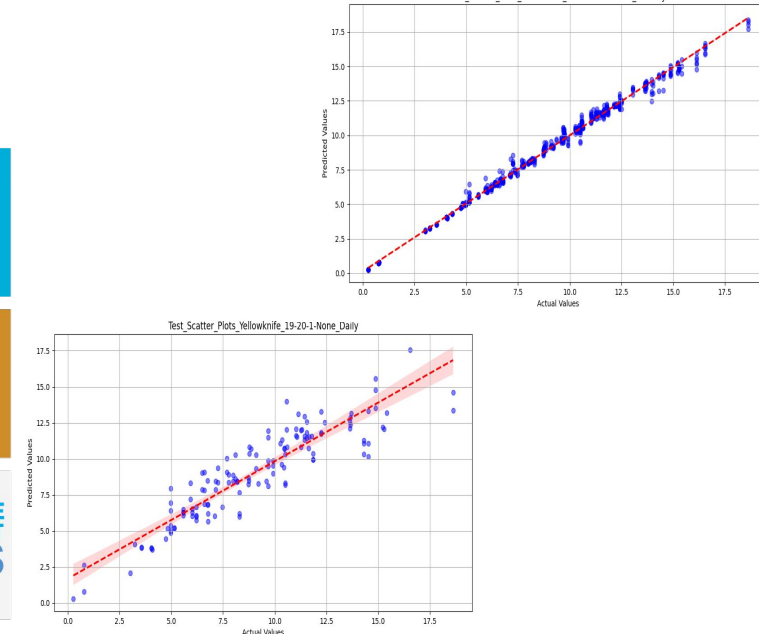
Forests

Forest areas management, carbon stock, potential fire areas detection, biomass estimation, forest floor status.

→ SDGs: 13, 15



SUSTAINABLE DEVELOPMENT GOALS



Natural Resources and Climate Change

Natural resources management, flood and drought mapping, rainfall prediction, energy management, solar calculator, heat islands in cities, human resources management, archaeological sites.

→ SDGs: 7 (Affordable & Clean Energy), 11 (Sustainable Cities), 13, 15

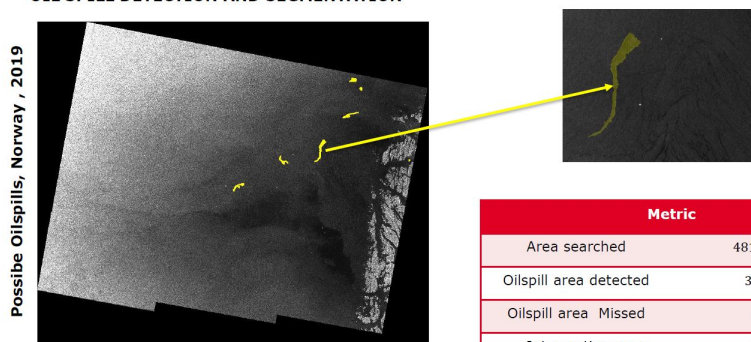


Maritime, Hydrologic

Vessel categorization and detection, oil spills, ice extension, water salinity, water turbidity, chlorophyll quantification, shoreline measurement, biodiversity, oceans and inland waters monitoring, aquifer monitoring, water quality measurement and monitoring.

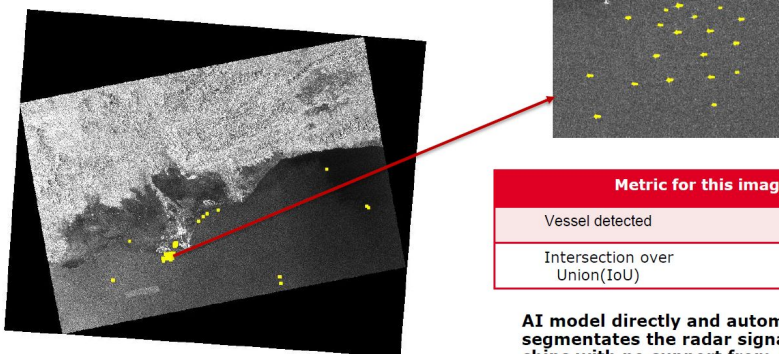
→ SDGs: 6 (Clean Water & Sanitation), 13, 14 (Life Below Water), 15

OIL SPILL DETECTION AND SEGMENTATION



Metric	
Area searched	48167km ²
Oilspill area detected	33km ²
Oilspill area Missed	4km ²
Intersection over Union(IoU)	0.88

VESSEL DETECTION AND SEGMENTATION



Metric for this image	
Vessel detected	0.98
Intersection over Union(IoU)	0.94

AI model directly and automatically segmentates the radar signature of ships with no support from operator

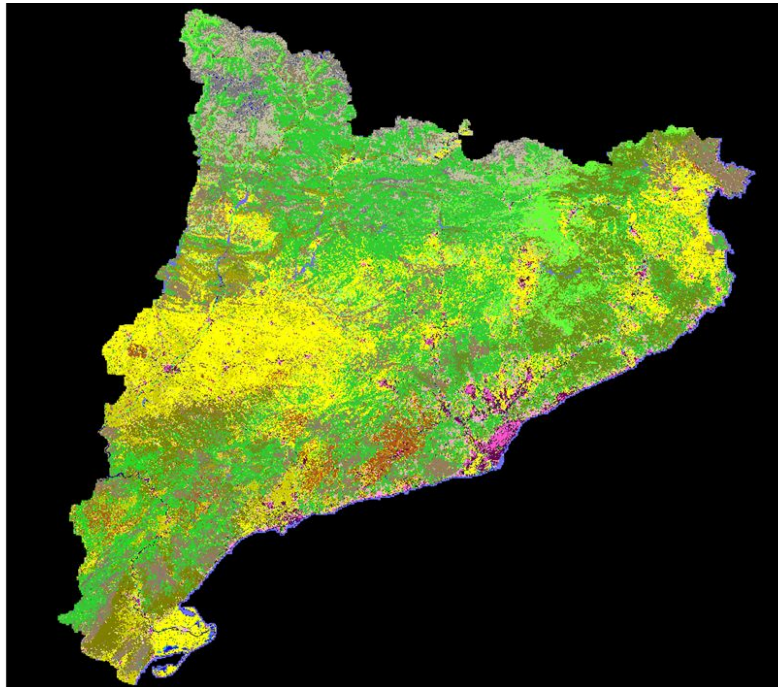
SUSTAINABLE DEVELOPMENT GOALS



Urban

Presence and distinction of buildings, detection of power lines and other objects, detection and prediction of population in rural areas, urban development, "Smart Cities" issues, detection of airplanes, light pollution, detection of illegal wells.

→ SDGs: 7, 9 (Industry, Innovation & Infrastructure), 11, 13



Atmosphere

Atmospheric monitoring, cloud detection, constellation monitoring, air particle detection.

→ SDGs: 3 (Good Health & Well-being), 11, 13



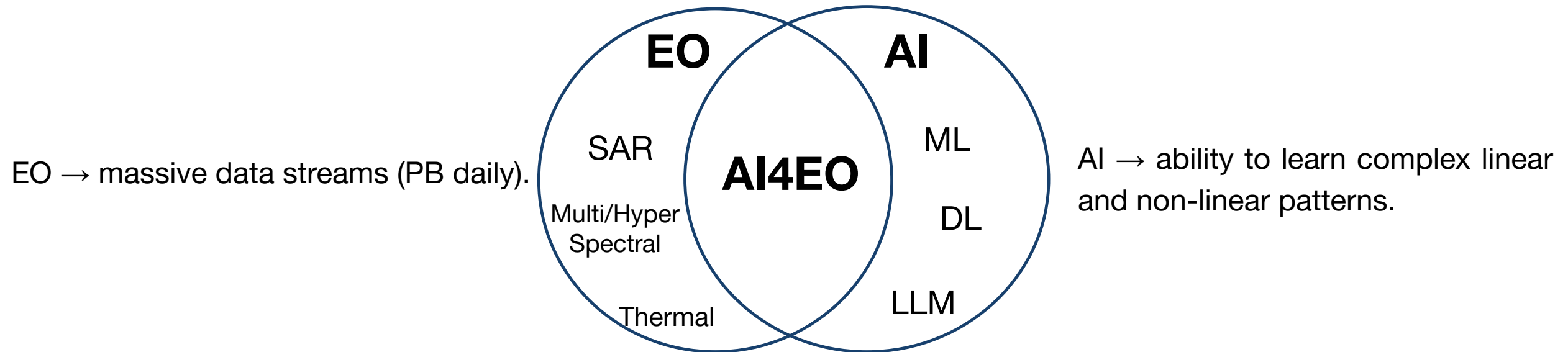
Security

Video forecasting, emergency services and management, security services, natural disasters management.

→ SDGs: 3, 11, 13, 16 (Peace, Justice & Strong Institutions)



- Increase in **computational power** → GPUs, TPUs, supercomputers.
- **Cloud** infrastructures → scalable storage & processing (AWS, Google Earth Engine, DIAS for Copernicus).
- Larger **storage** capacities → handling petabyte-scale EO archives.
- Lower **processing** times → from weeks/months to near-real time.
- “New” larger AI models → Deep Learning (CNNs, Transformers), **LLMs** applied to EO data.
- New opportunities → multimodal fusion (EO + **IoT** + socioeconomic data).



AI4EO: turning data into **actionable intelligence**.



AI4EO in Catalonia

AI4EO Sector in Catalonia



1. Select a **topic/use case** and define the **problem** to solve.

Considerations:

- Scope → “What does it cover?”
- Impact → “Why does it matter?”
- Vision → “Where is it going?”
- Objectives → “What will be achieved?”
- Limitations → “What can/not be done?”
- Business → “Who will use or benefit from it?”

Term 1: Definition

2. Define the **initial boundary conditions**: temporal and spatial extent, other conditions.
3. **Data available**, both EO and Ground Reference. Open Access, preprocess required.
4. Define the **final boundary conditions**: temporal and spatial extent, other conditions.

Plan and account for everything from the start

Term 2: Implementation

1. Radar (SAR – Synthetic Aperture Radar, Altimetry)

- **SAR:** active sensor that sends radar pulses → can "see" through clouds and at night.
- **Altimetry:** measures surface height (oceans, ice sheets, land).
- **Use cases:** flood mapping, soil moisture, ship detection, sea level monitoring.

2. Multi- & Hyperspectral

- **Multispectral:** captures a few broad bands (e.g., Sentinel-2 with 13 bands).
- **Hyperspectral:** captures hundreds of narrow bands → detailed spectral signatures.
- **Use cases:** agriculture, forestry, minerals, pollution detection.

3. Thermal Infrared

- Measures heat emitted from surfaces (day & night).
- **Use cases:** urban heat islands, wildfire detection, water stress in crops.
- **Main provider:** NASA LP DAAC (Land Processes Distributed Active Archive Center) (e.g., MODIS, ECOSTRESS).

4. Other Sensors

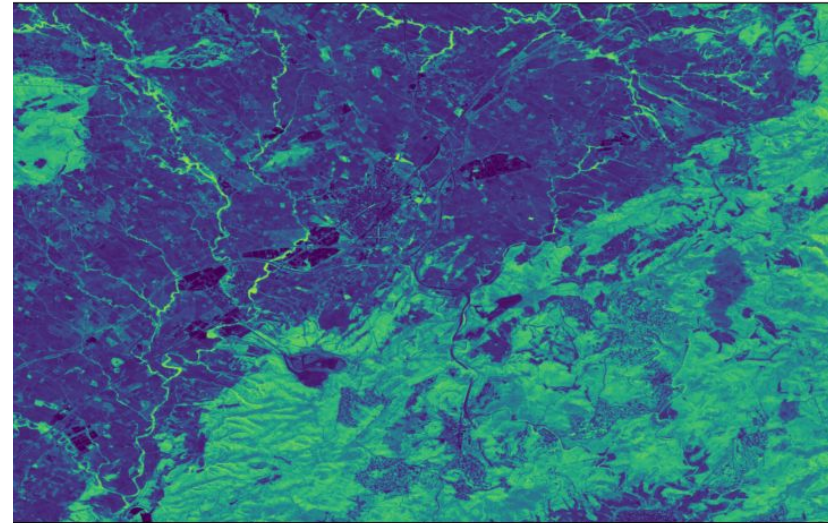
- **LiDAR:** laser pulses → precise elevation maps
- **GNSS-Reflectometry:** uses GPS signals reflected from Earth → soil moisture, ocean roughness

Earth Observation Data

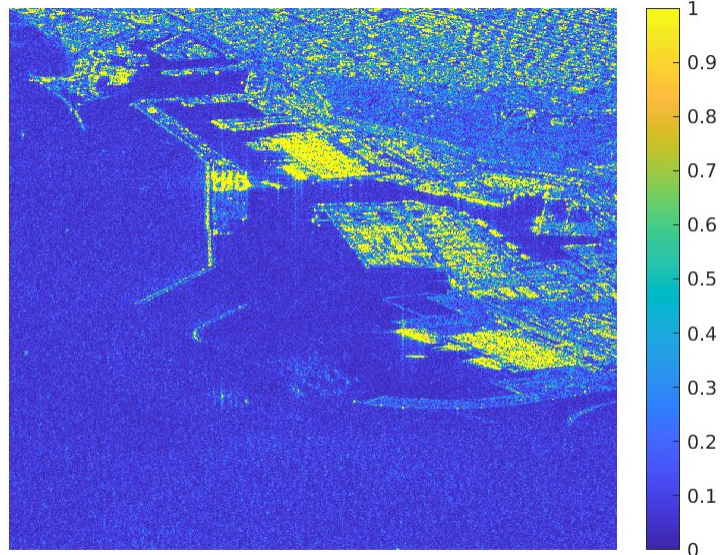
MultiSpectral (RGB) 10m



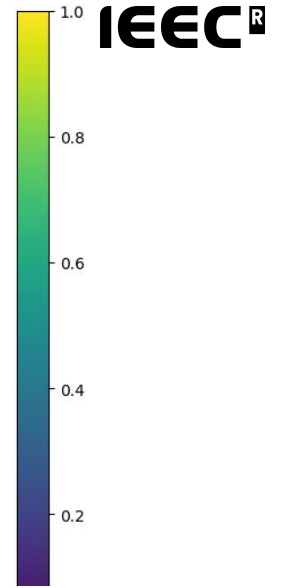
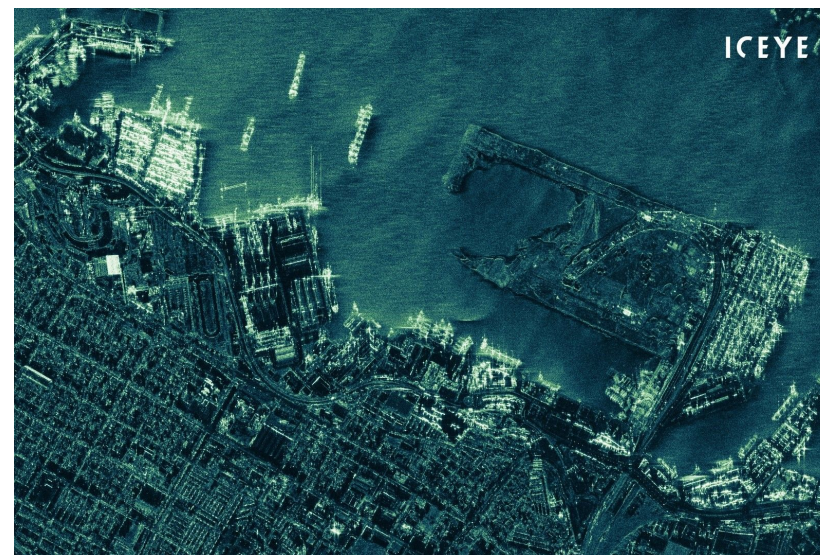
MultiSpectral 10m



SAR 10m



SAR 0.5m



Data Type	What it Measures	Example Use Cases	Main Sources
Radar (SAR)	Surface structure, moisture, movement (all-weather, day/night)	Flood mapping, deforestation, ship detection	Copernicus Sentinel-1
Radar (Altimetry)	Surface height (oceans, land, ice)	Sea level rise, ice sheet monitoring	ESA CryoSat
Multispectral	Reflectance in a few broad bands	Agriculture, land cover, vegetation health	Copernicus Sentinel-2
Hyperspectral	Reflectance in hundreds of narrow bands	Mineral exploration, crop stress, pollution	NASA AVIRIS
Thermal Infrared	Heat emitted from surfaces	Wildfires, water stress, urban heat islands	NASA LP DAAC (MODIS, ECOSTRESS)

- Each data type → different information layer.
- Combining them = **multimodal AI models** → richer insights (e.g., combining SAR + optical for deforestation, or thermal + hyperspectral for precision agriculture).

What it is:

- Ground Reference Data = direct measurements taken on Earth to calibrate/validate EO satellite data.
- Essential for training & validating AI models → ensures accuracy and real-world reliability.

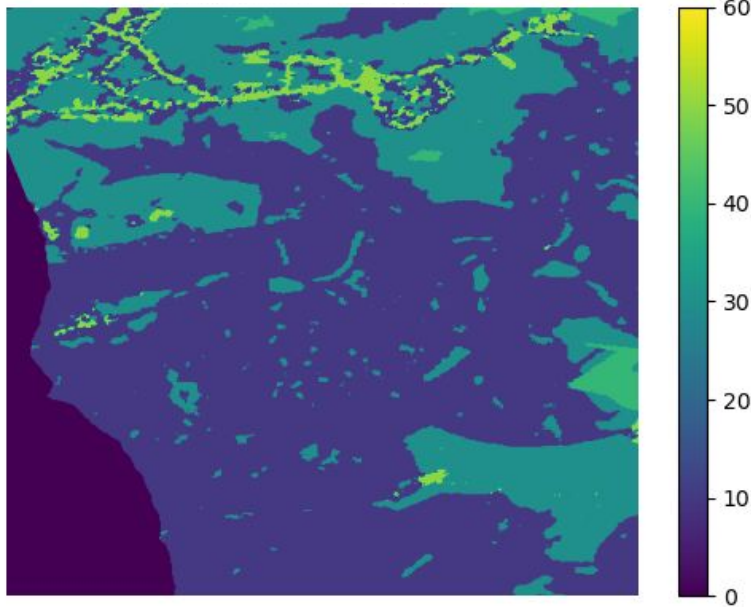
Examples:

- Biomass → tree inventories, lidar field campaigns.
- Land Cover → classification maps from surveys.
- DEM (Digital Elevation Models) → terrain height maps (e.g., SRTM, Copernicus DEM).
- Meteorology → ground weather stations (temp, rainfall, wind).
- In-situ campaigns → soil samples, crop data, water quality, manual measurements.

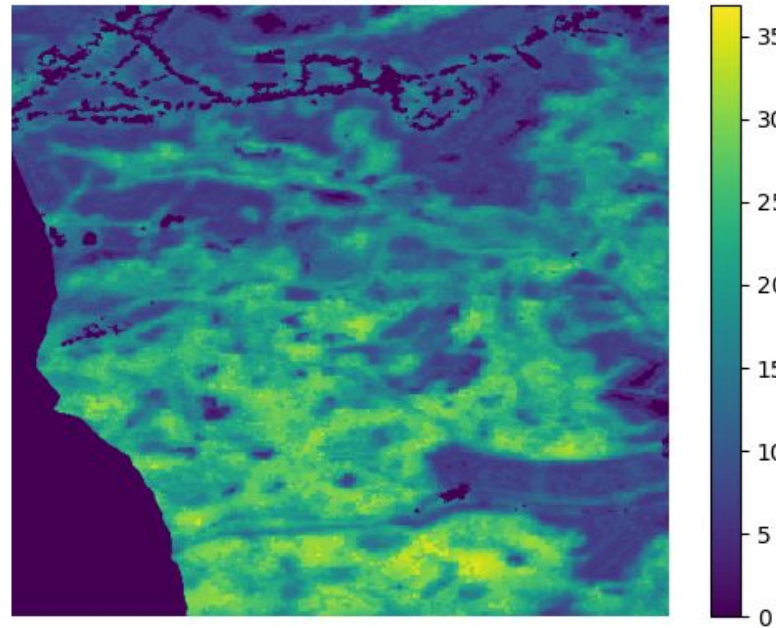
Who provides it:

- ESA – European Space Agency → [ESA Earth Observation Data](#)
- NASA – Earth Science Data Systems (ESDS) → [NASA Earthdata](#)
- USGS – U.S. Geological Survey → [USGS Earth Explorer](#)
- FAO (Food and Agriculture Organization, UN) → [FAO GeoNetwork](#)
- ECMWF – European Centre for Medium-Range Weather Forecasts (ERA5 reanalysis) → [ERA5 Data via Copernicus Climate Data Store](#)

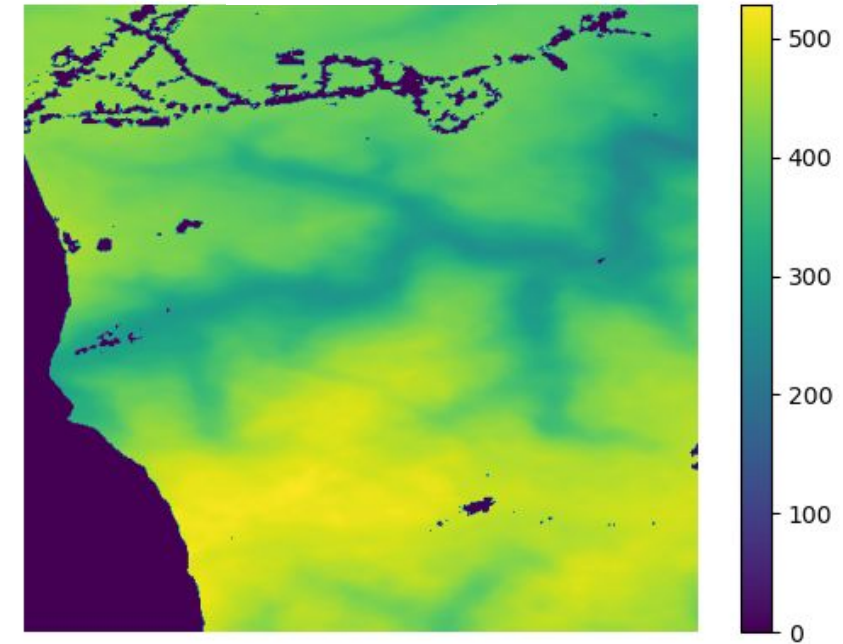
Land Cover (LC)



Tree Height (TH)



DEM



Complex Problem Solving

1-2-4-all

Write down **2 use cases** that you think **a specific group of people** may benefit from or fulfil a **necessity**: in a specific place, a specific solution.

Write them as SOLUTION | PLACE | COMMUNITY

e.i. animals monitoring | mountain regions | ranchers

Time per round 4 minutes (2min/idea or person or group).

Round	# People	# Use Cases
1	1	2
2	2	3
3	4	4
4	8	5
5	All	-